

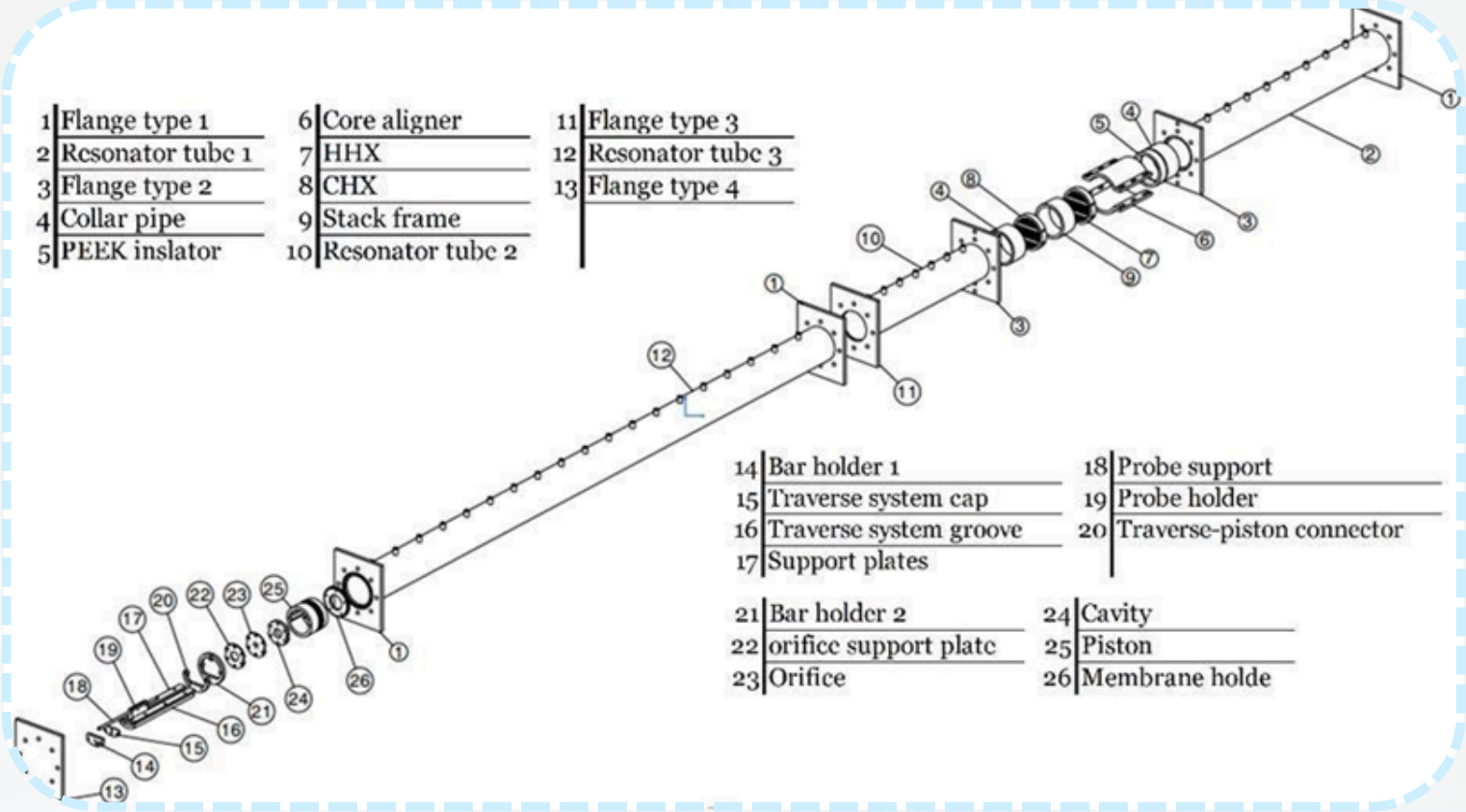
VORTEK

“Silent by nature, sustainable by design.
Acoustic waves, vortex jets, no compromise.”

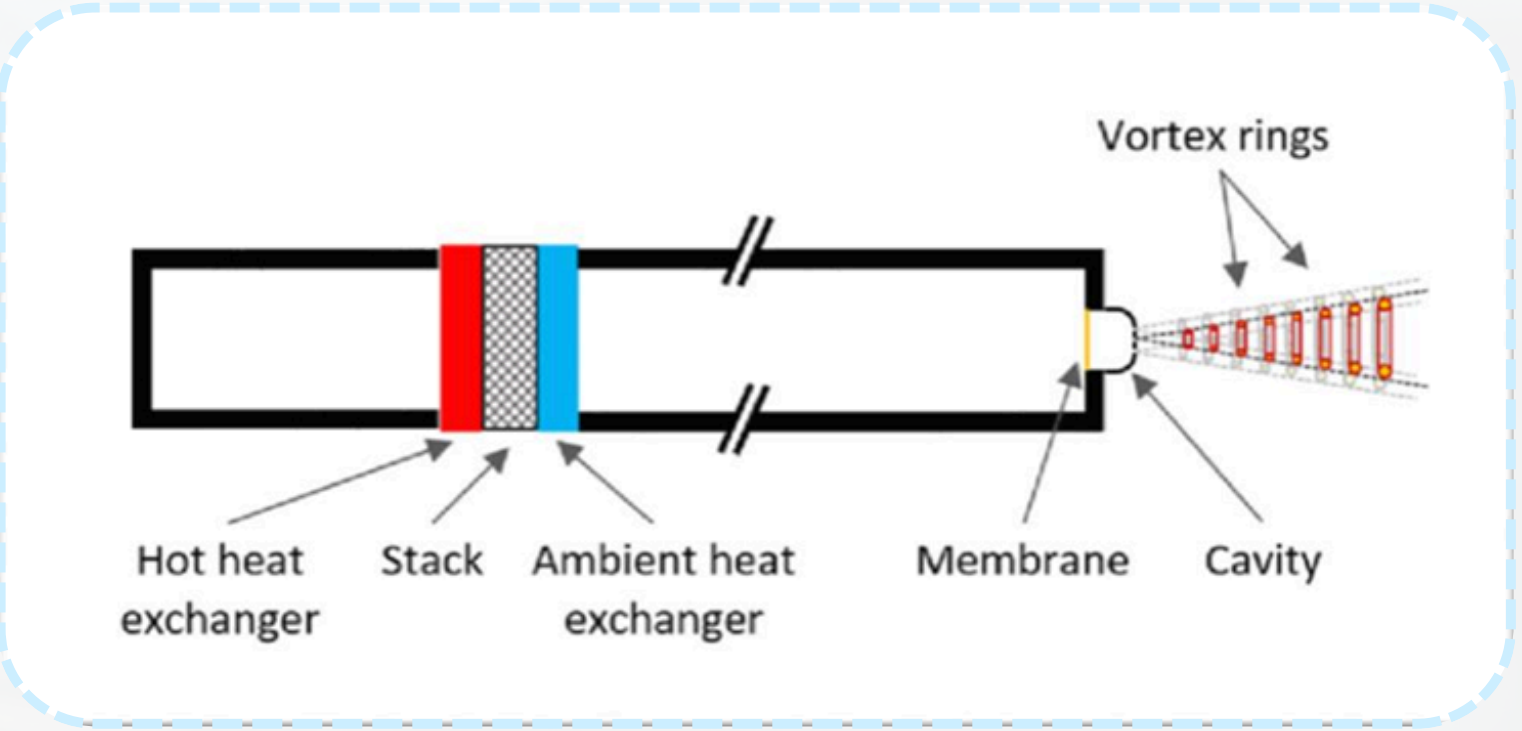


MISSION & ROLE :

- **VORTEK prototype:** a 26 mm synthetic-jet actuator (SJA) driven only by a thermoacoustic engine's (TAE) acoustic wave – no fans, no refrigerant.
- **My role:** full CAD, resonator sizing, SJA orifice & diaphragm design, data acquisition (LabVIEW + MATLAB), and analysis.
- **Goal:** deliver silent (< 35 dBA), refrigerant-free cooling for dense electronics by recycling waste heat into pulsed jets.



Schematic Diagram of TAE and SJA



Integrated TAE and SJA System

	1	2	3	4	5	6	7	8	9	10	11	12	13
Xs	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%
Xhs	460	455	450	445	440	435	430	425	420	415	410	405	400
Xp2-Hot side	87	92	97	102	107	112	117	122	127	132	137	142	147
Total Length	870	920	970	1020	1070	1120	1170	1220	1270	1320	1370	1420	1470
Xp1-cold side	783	828	873	918	963	1008	1053	1098	1143	1188	1233	1278	1323
Xcs	746	701	656	611	566	521	476	431	386	341	296	251	206
f _{req}	198.85	188.04	178.35	169.61	161.68	154.46	147.86	141.80	136.22	131.06	126.28	121.83	117.69
	420	415	410	405	400	395	390	385	380	375	370	365	360
	557	512	467	422	377	332	287	242	197	152	107	62	17
P4 location/Lr %	23.79	23.04	22.37	21.76	21.21	20.71	20.26	19.84	19.45	19.09	18.76	18.45	18.16
P3 location/Lr %	41.26	39.57	38.04	36.67	35.42	34.29	33.25	32.30	31.42	30.61	29.85	29.15	28.50
P2 location/Lr %	77.70	74.02	70.72	67.75	65.05	62.59	60.34	58.28	56.38	54.62	52.99	51.48	50.00
P1 location/Lr %	100	100	100	100	100	100	100	100	100	100	100	100	100

Experiment Configuration for Different Test



Synthetic Jet Actuator

CONCEPT & INNOVATION:

- Thermoacoustic engine
 ΔT drives 96 dB acoustic wave
- Synthetic-jet actuator
 Transforms pressure into coherent vortex rings
- Resonator tuning
 Four lengths tested: 1020–1420 mm; find sweet spots
- Orifice design
 1.2mm star orifice + 0.2mm latex diaphragm for rapid jet pulses

EXPERIMENTAL RIG & INSTRUMENTATION:

Heat source

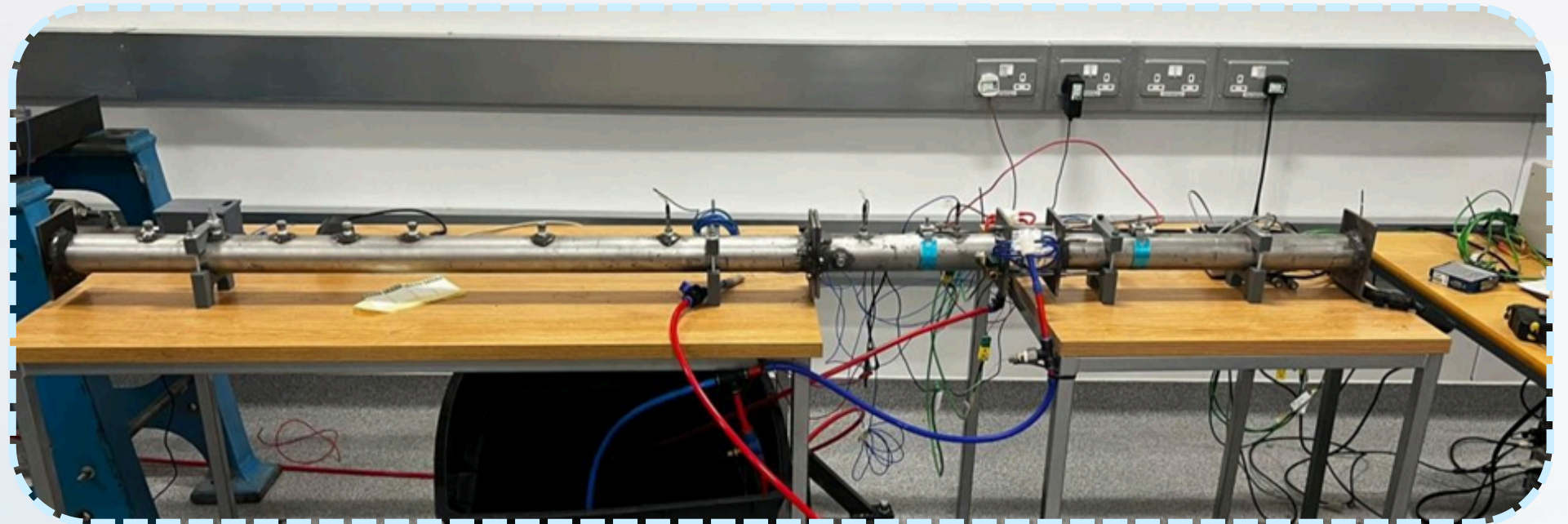
DC Variac heats HHX to 338 °C,
stabilizes at 115 V

Signal processing

MATLAB 100-pt moving average
filter for clean FFT analysis

Data acquisition

Pressure sensors at 4.1 kHz, hot-
wire at 4 kHz, 32,769
samples/run



Experiment Rig Setup

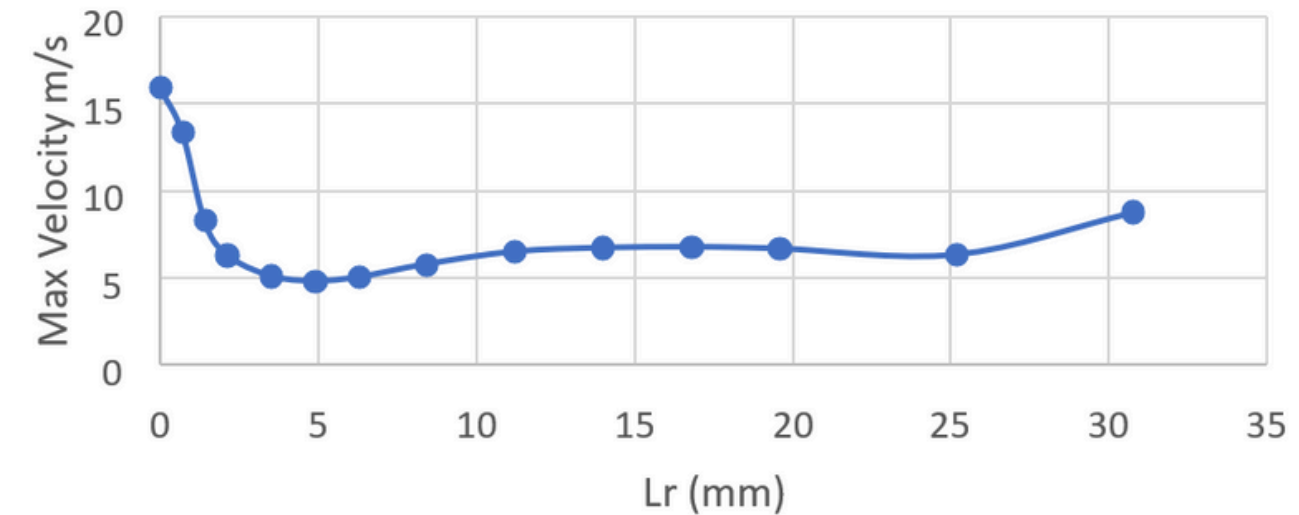


Hot Wire attached to the Probe Holder

FLOW FIELD RESULTS:

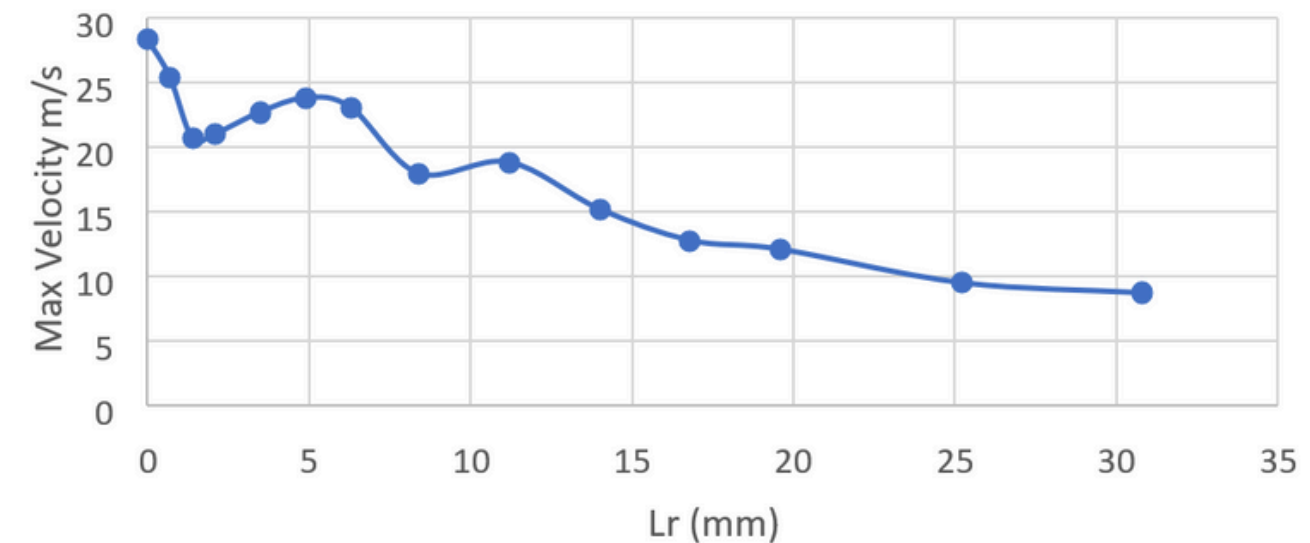
- **Peak jet 30 m s^{-1}** at nozzle; velocity plateaus **6–8 m s^{-1}** from **15–35 mm**, giving a useful cooling reach.
- Rapid decay within first **5 mm** highlights need to position **heat-sinks close to nozzle** for **maximal heat flux**.
- **Stable plateau region** indicates sustained convective enhancement with minimal turbulence noise.

Avg Velocity Vs Lr for T-10



Avg Velocity Vs HW Distance for T-10

Max Velocity Decay for T-10



Max Velocity Vs HW Distance for T

THERMAL & ACOUSTIC PERFORMANCE



Optimal Resonator Length

1,150 mm yields 338 °C ΔT across heat exchangers



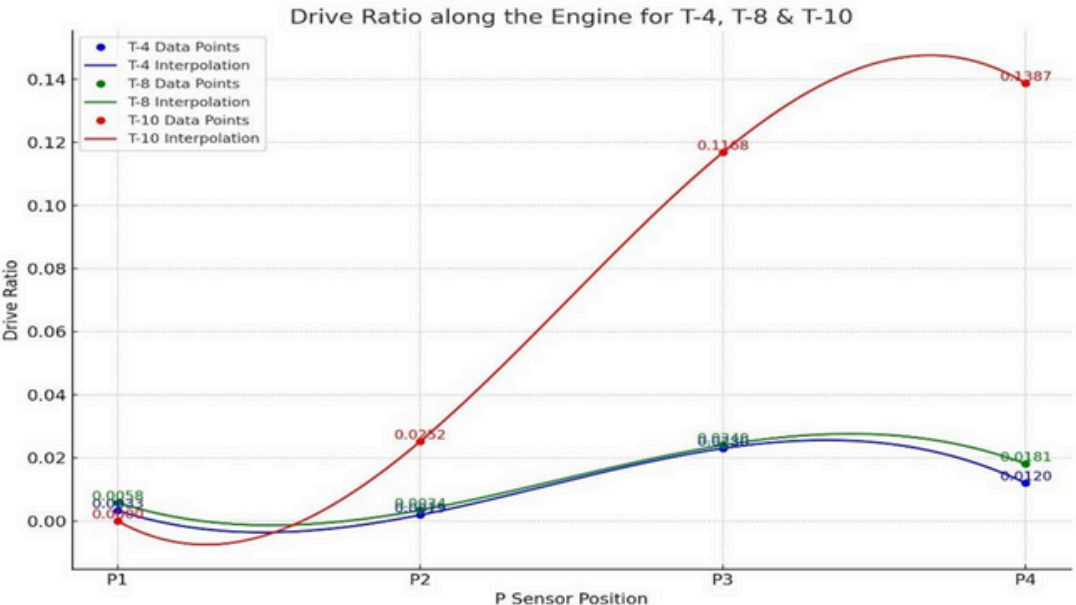
Drive Ratio Peak

0.116–0.137 near pressure antinode confirms efficiency

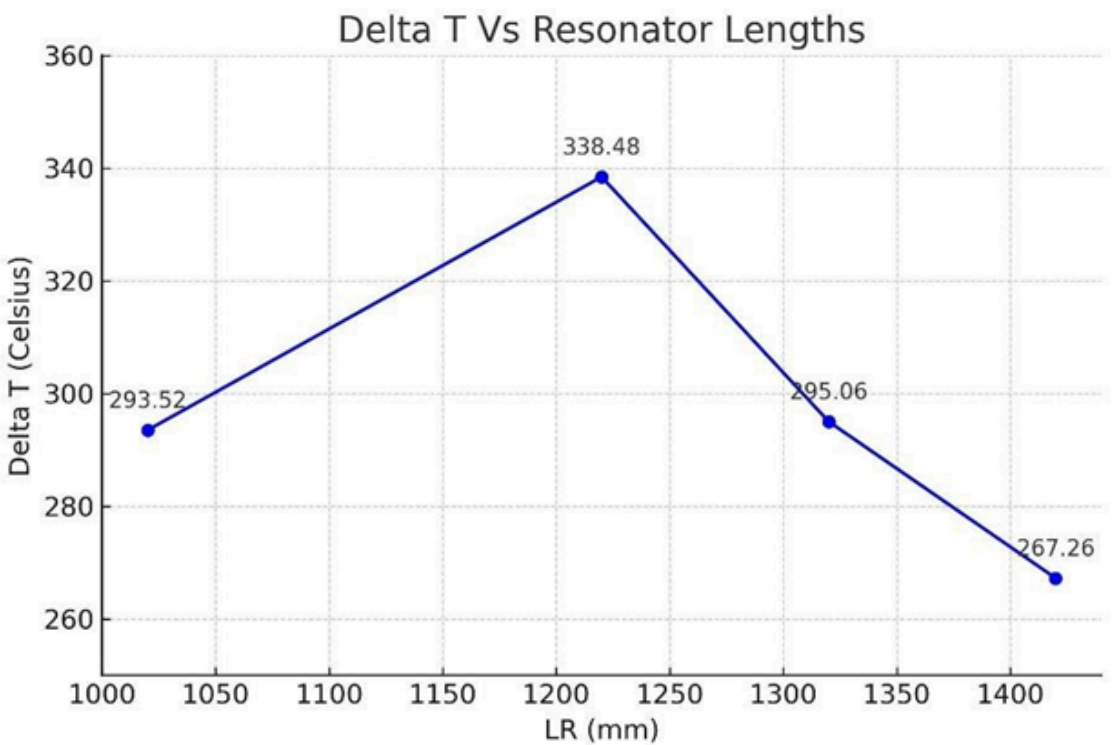


Silent Operation

Air pulses > 18 m/s with zero fan power, under 35 dBA



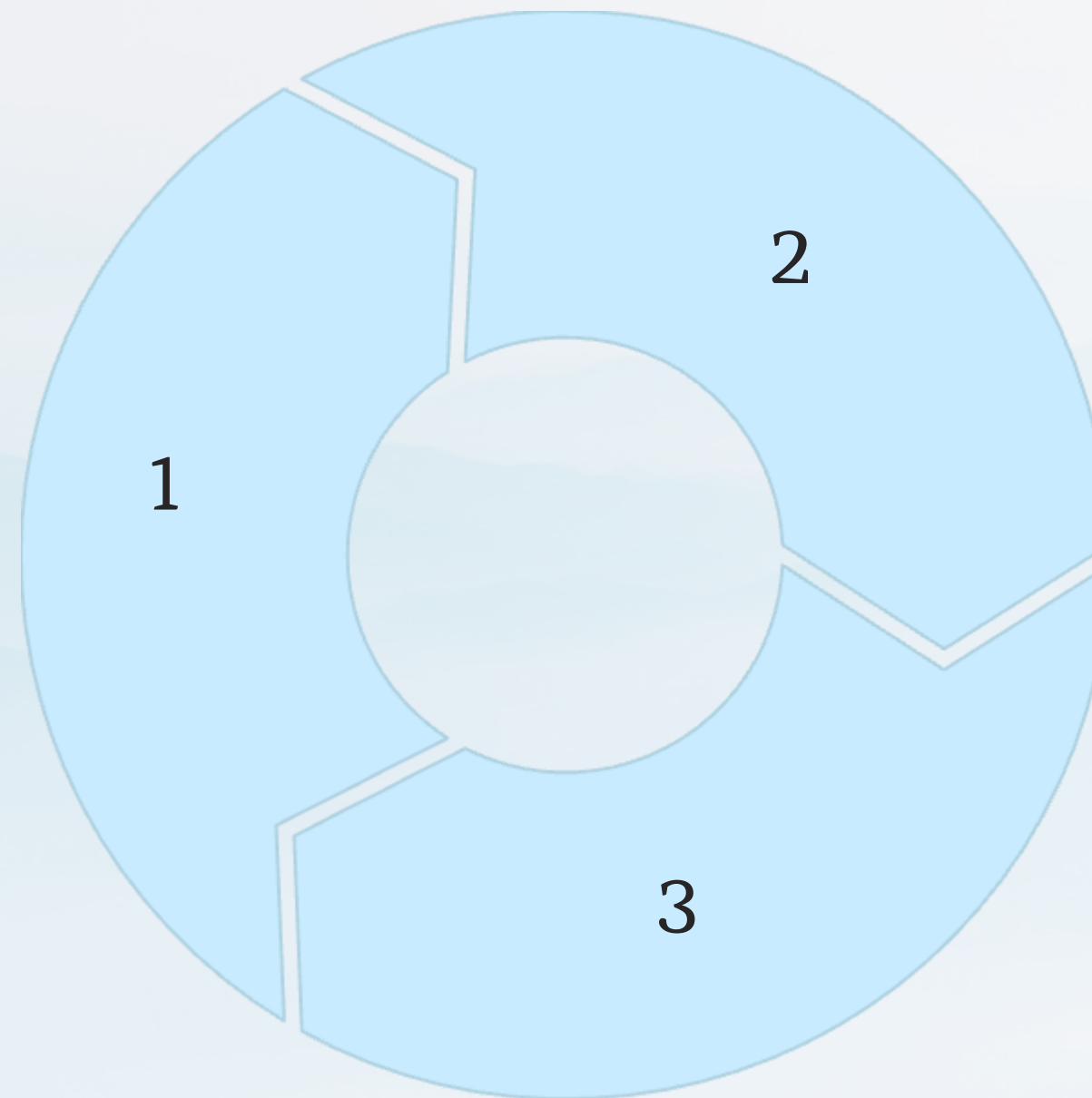
Combined Graph of P Sensor Location Vs DR



Delta T vs Resonator Lengths Graph for T-4, T-8,T-10 and T-12

IMPACT & APPLICATIONS:

Eco-friendly Cooling
Eliminates refrigerants and fans



Scalable Design

Tunable resonator for gadgets to
100kW industrial use

Green Tech Alignment

Maintenance-free, long-life thermal
management solution